COMP 3981: Project Description (Part #3)

Search Strategy and Performance Enhancements

General Instruction:

For this part of the project, you will work with your team members on completing the design and implementation for the abalone game-playing agent in preparation for tournament.

I Game-playing Agent

The game-playing logic of the agent must include:

- minimax algorithm with alpha-beta pruning
- heuristic evaluation function for real-time decisions
- first move (for the black player) be a random legal move
- integration with Part #1 and Part #2 deliverables

In addition, you must also incorporate at least one of the following performance enhancements to your game-playing agent (you may expect the per player, per-move time limit can be as small as 5 seconds):

- node ordering
- transposition tables
- quiescence search

II Heuristic Evaluation Function

The performance of a game-playing agent depends strongly on the quality of its heuristic evaluation function. Now that you have had the opportunity to brainstorm and discuss evaluation functions in general with your team members, you are required to individually:

i. design and implement a heuristic evaluation function
ii. evaluate the performance of your heuristic evaluation function
iii. convince your team members to adopt your heuristic evaluation function for your team's game-playing agent

BONUS: The team member, whose heuristic evaluation function is adopted for the team's game-playing agent, will be awarded an individual 10% bonus project mark. Note that the bonus mark will only be awarded to one (1) team member per team and is not dividable among team members. Hence, it is imperative that you rigorously convince your team members, based on performance evaluation, to adopt your heuristic evaluation function for your team's game-playing agent. In the event that decision unanimity is not reached within a team, the bonus for that team is forfeited.

Deliverables:

All work submitted is subject to the standards of conduct as specified in BCIT Policy 5104. For all deliverables, late submissions will not be accepted and you are allowed only one submission.

You are expected to deliver a fully functional abalone game-playing agent as specified in (I) for Part #3. A round-robin trial tournament will be held in-class for this part of the project. Only minor bug-fixes, further performance enhancements, additional testing and final documentation should be done for Part #4.

The deliverables, one per group (unless otherwise indicated), for Part #3 (BCIT Learning Hub submission) are as follows:

1) [50 marks] [Mar 28, 2024 @2359] Submit your Search Strategy and Performance Enhancement report (for Part #3) which must include, at a minimum, the following:
   a. Design and architecture of your game-playing agent
   b. Detailed description of the adopted heuristic evaluation function (including how the heuristic evaluation function was evaluated as well as performance results)
   c. Detailed description of all the performance enhancement(s) incorporated to your game-playing agent
   d. Clearly explain and justify the choice of values if your heuristic evaluation function in (1-b) and/or performance enhancement(s) in (1-c) make(s) use of any constants/weights
   e. Name of the team member whose heuristic evaluation function was adopted (including justification for the decision)
   f. Team member contribution
   g. List of references used in the development of your abalone game-playing agent
   h. Any additional documentation (as determined by each team) pertinent to the assessment of your deliverable

   in ***part3report_groupnumber.pdf***, to BCIT Learning Hub (Project Submission | Part 3: Report).

2) [20 marks] [Mar 28, 2024 @2359] Submit, one per team member, in the appendix of the report in (1), a detailed description of the heuristic evaluation function proposed for (II) (including performance results as well as how the heuristic function was evaluated).
   (Note: If your heuristic evaluation function was adopted and detailed in (1-b), you do not have to submit this individual write-up.)

3) [30 marks] [Mar 28, 2024 @2359] Submit your source code and executable including detailed instructions on how to build and run the executable in ***part3sourcecode_groupnumber.zip***, to BCIT Learning Hub (Project Submission | Part 3: Source Code). You are allowed only one submission.
(Note: You will receive a mark of zero (i) if your code does not compile or execute on the (freshly-imaged) instructor terminal in the option lab (SE12-306) or (ii) you do not provide detailed instructions on how to (build and) run the executable.)

Note that there is no in-class demonstration for this part of the project.

Upcoming Project Due Dates:

Apr 4, 2024: Project Part #4
Tournament:
Agent Round-robin Trials

Apr 4, 2024: Project Part #4
Tournament:
Agent Heats, Semi-final, Final
Agent Champion vs. Human Player (time-permitting)
Final Deliverables Submission